

# Adaptive denoising diffusion modelling via random time reversal

ICSDS Sevilla

---

Lukas Trottnner

based on joint work with [Sören Christensen](#), [Jan Kallsen](#) and [Claudia Strauch](#)

15 December 2025

University of Stuttgart

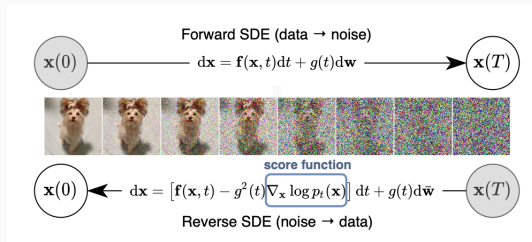
[Kiel University](#)    [Heidelberg University](#)



**University of Stuttgart**  
Germany

# Denoising diffusion models

- provide an **iterative generative algorithm** to create new samples that approximately match the target distribution  $p_0$ , given a finite number of samples corresponding to an unknown  $p_0$
- general idea: find a **stochastic process** that perturbs  $p_0$  to a new distribution  $p_T$  such that
  - 1)  $p_T$  or a good approximation thereof is **easy to sample from**, and
  - 2) the perturbation is **reversible** in the sense that we know how to **simulate the time-reversed process**



Source: Song et al. (2021). Score based generative modeling through stochastic differential equations. *ICLR*.

# Denoising Diffusion Models

- for some fixed time  $T > 0$  consider the forward model

$$dX_t = b(t, X_t) dt + \sigma(t, X_t) dW_t, \quad t \in [0, T], X_0 \sim p_0$$

# Denoising Diffusion Models

- for some fixed time  $T > 0$  consider the **forward model**

$$dX_t = b(t, X_t) dt + \sigma(t, X_t) dW_t, \quad t \in [0, T], X_0 \sim p_0$$

- under sufficient regularity conditions, the forward model has a solution  $X = (X_t)_{t \in [0, T]}$  with marginal densities  $p_t(x) = \int p_{0,t}(y, x) p_0(dy)$  such that the **time reversal**  $\tilde{X}_t = X_{T-t}$  solves

$$d\tilde{X}_t = -\bar{b}(T-t, \tilde{X}_t) dt + \sigma(T-t, \tilde{X}_t) d\bar{W}_t, \quad t \in [0, T], \tilde{X}_0 \sim p_T,$$

where

$$\begin{aligned} \bar{b}_i(t, x) &= b_i(t, x) - \frac{1}{p_t(x)} \sum_{j,k=1}^d \frac{\partial}{\partial x_j} (p_t(x) \sigma_{ik}(t, x) \sigma_{jk}(t, x)) \\ &= b_i(t, x) - (\nabla \cdot \Sigma(t, x))_i - (\nabla \log p_t(x))_i, \quad i = 1, \dots, d, \Sigma = \sigma \sigma^\top \end{aligned}$$

# Denoising Diffusion Models

- for some fixed time  $T > 0$  consider the **forward model**

$$dX_t = b(t, X_t) dt + \sigma(t, X_t) dW_t, \quad t \in [0, T], X_0 \sim p_0$$

- under sufficient regularity conditions, the forward model has a solution  $X = (X_t)_{t \in [0, T]}$  with marginal densities  $p_t(x) = \int p_{0,t}(y, x) p_0(dy)$  such that the **time reversal**  $\tilde{X}_t = X_{T-t}$  solves

$$d\tilde{X}_t = -\bar{b}(T-t, \tilde{X}_t) dt + \sigma(T-t, \tilde{X}_t) d\bar{W}_t, \quad t \in [0, T], \tilde{X}_0 \sim p_T,$$

where

$$\begin{aligned} \bar{b}_i(t, x) &= b_i(t, x) - \frac{1}{p_t(x)} \sum_{j,k=1}^d \frac{\partial}{\partial x_j} (p_t(x) \sigma_{ik}(t, x) \sigma_{jk}(t, x)) \\ &= b_i(t, x) - (\nabla \cdot \Sigma(t, x))_i - (\nabla \log p_t(x))_i, \quad i = 1, \dots, d, \Sigma = \sigma \sigma^\top \end{aligned}$$

- ~> time-reversed process solves a **time-inhomogeneous SDE**, now with drift  $-\bar{b}(T - \cdot, \cdot)$  involving the **score**  $\nabla \log p_t$ , which depends on the **unknown** data distribution  $p_0$
- ~> score needs to be estimated from the data

## Generative process

- given data  $(X_0^i)_{i \in [n]} \stackrel{\text{iid}}{\sim} p_0$  define the **denoising score estimator**

$$\hat{\mathfrak{s}} \in \arg \min_{s \in \mathcal{S}} \frac{1}{n} \sum_{i=1}^n \mathbb{E}_{X_0^i} \left[ \int_{\underline{T}}^T \|s(t, X_t) - \nabla_2 \log p_{0,t}(X_0, X_t)\|^2 dt \right],$$

- On  $[0, T - \underline{T}]$ , simulate

$$dY_t = \left( -b(T-t, Y_t) + \nabla \cdot \Sigma(T-t, Y_t) + \Sigma(T-t, Y_t) \hat{\mathfrak{s}}(T-t, Y_t) \right) dt + \sigma(T-t, Y_t) dW_t, \quad \mathbb{P}^{Y_0}(dy) \approx p_T(y) dy$$

- Output:  $Y_{T-\underline{T}} \stackrel{d}{\approx} \tilde{X}_{T-\underline{T}} = X_{\underline{T}} \stackrel{d}{\approx} X_0$

---

<sup>1</sup>Stanczuk et al. (2024). Your diffusion model secretly knows the dimension of the data manifold. *ICML*.

## Generative process

- given data  $(X_0^i)_{i \in [n]} \stackrel{\text{iid}}{\sim} p_0$  define the **denoising score estimator**

$$\hat{s} \in \arg \min_{s \in \mathcal{S}} \frac{1}{n} \sum_{i=1}^n \mathbb{E}_{X_0^i} \left[ \int_{\underline{T}}^T \|s(t, X_t) - \nabla_2 \log p_{0,t}(X_0, X_t)\|^2 dt \right],$$

- On  $[0, T - \underline{T}]$ , simulate

$$dY_t = \left( -b(T-t, Y_t) + \nabla \cdot \Sigma(T-t, Y_t) + \Sigma(T-t, Y_t) \hat{s}(T-t, Y_t) \right) dt + \sigma(T-t, Y_t) dW_t, \quad \mathbb{P}^{Y_0}(dy) \approx p_T(y) dy$$

- Output:  $Y_{T-\underline{T}} \stackrel{d}{\approx} \tilde{X}_{T-\underline{T}} = X_{\underline{T}} \stackrel{d}{\approx} X_0$

### Basic observations

- time reversal at **deterministic** time  $T$  forces the backward process to be time-inhomogeneous
- if  $p_0$  has **low-dimensional support**  $\mathcal{M}$ , for small  $t$  and  $x$  close to  $\mathcal{M}$ ,  $\nabla \log p_t(x)$  is approximately **orthogonal** to  $\mathcal{M}$  (Stanczuk et al., 2024)<sup>1</sup>
- algorithm is **not adaptive** to the noise level in the data

---

<sup>1</sup>Stanczuk et al. (2024). Your diffusion model secretly knows the dimension of the data manifold. *ICML*.

## $h$ -transforms and time reversal

### $h$ -transform

For a possibly killed, homogeneous strong Markov process  $X$  with state space  $S$ , let  $h$  be an excessive function, that is

$$\mathbb{E}_x[h(X_t)] \leq h(x) \quad \text{and} \quad \lim_{t \rightarrow 0} \mathbb{E}_x[h(X_t)] = h(x).$$

Then,

$$p_t^h f(x) = \mathbb{E}_x \left[ \frac{h(X_t)}{h(x)} f(X_t) \mathbf{1}_{\{X_t \in S\}} \right] \mathbf{1}_{(0, \infty)}(h(x)), \quad f \in \mathcal{B}_b(\mathbb{R}^d),$$

defines a sub-Markov semigroup. The corresponding Markov process  $X^h$  is strong Markov and is called  $h$ -transform of  $X$ .

- suppose that  $X$  is a continuous and [self-dual](#) Feller process (i.e., its generator satisfies  $A = A^*$ )
- if  $X^h$  has a finite killing time  $\zeta$ , then the time-reversed process  $\bar{X}_t^h = X_{\zeta-t}^h$  is [homogeneous, strong Markov](#) and is a  $\bar{h}$ -transform of  $X$ .



## $h$ -transforming a killed diffusion

- consider a **symmetric** diffusion process

$$dX_t = b(X_t) dt + \sigma(X_t) dW_t$$

with invariant measure  $m$  and let  $Z$  be its version **killed at an independent exponential time** with parameter  $r > 0$

- as an excessive function for  $Z$  use

$$h(x) = \int G_r(x, y) \kappa(dy)$$

for the **Green kernel**  $G_r(x, y) = \int_0^\infty e^{-rt} p_t(x, y) dt$  and a **representing measure**  $\kappa$

- $\kappa(dy) = r dy \rightsquigarrow h = 1$  and  $Z^h = Z$
- $\kappa(dy) = \frac{1}{G_r(x_0, y)} \beta(dy) \rightsquigarrow Z$  conditioned to have distribution  $\beta$  before killing if started in  $x_0$
- $Z$  is a killed Brownian motion and  $\kappa(dy) = \sigma_R(dy)$  for the surface measure  $\sigma_R$  of an  $R$ -sphere  $\mathbb{S}^{d-1}(R) \rightsquigarrow Z^h$  is killed at last exit from  $\mathbb{S}^{d-1}(R)$

# A time-homogeneous generative process

## Proposition

1.  $Z^h$  is an Itô-diffusion with dynamics

$$dZ_t^h = (b(Z_t^h) + \Sigma(X_t) \nabla \log h(X_t)) dt + \sigma(Z_t^h) dW_t$$

outside  $\text{supp } \kappa$  and its distribution at the lifetime is given by

$$\mathbb{P}_x(Z_{\zeta_-}^h \in dy) = \frac{G_r(x, y)}{h(x)} \kappa(dy)$$

2. Let  $\alpha = \mathbb{P}^{Z_0^h}$ . Then  $\tilde{Z}_t^h$  is an  $\tilde{h}$ -transform of  $Z$  with initial distribution  $\mathbb{P}_\alpha(Z_{\zeta_-}^h \in dy)$  and

$$\tilde{h}(x) := \int \frac{G_r(x, y)}{h(y)} \alpha(dy).$$

In particular,  $\tilde{Z}^h$  has dynamics

$$d\tilde{Z}_t^h = (b(\tilde{Z}_t^h) + \Sigma(\tilde{Z}_t^h) \nabla \log \tilde{h}(\tilde{Z}_t^h)) dt + \sigma(\tilde{Z}_t^h) d\bar{W}_t,$$

outside  $\text{supp } \alpha =: \mathcal{M}$  and  $\mathbb{P}_\alpha(Z_{\zeta_-}^h \in dy \mid Z_0^h = x) = \frac{G_r(x, y)}{\tilde{h}(x)h(y)} \alpha(dy)$  for  $\mathbb{P}_\alpha(Z_{\zeta_-}^h \in \cdot)$ -a.e.  $x$ .

# A time-homogeneous generative process

Idealised algorithm:

1. Initialise  $Z_0^{\tilde{h}} \sim \tilde{\beta} \approx \mathbb{P}_\alpha(Z_{\zeta-}^h)$ 
  - for ergodic forward process with stationary distribution  $\mu$  and small exponential killing rate  $r > 0$ , choose  $\tilde{\beta} = \mu$  [ $\leftrightarrow$  [ergodic diffusion model](#)]
  - for exponentially killed BM with small killing rate  $r > 0$ , choose  $\tilde{\beta} = \text{Laplace}(0, (2r)^{-1/2} \mathbb{I}_d)$  [ $\leftrightarrow$  [variance exploding diffusion model](#)]
  - for  $\kappa(dy) = \frac{1}{G_r(x_0, y)} \delta_z$ , choose  $\tilde{\beta} = \delta_z$
2. Simulate diffusion  $Z^{\tilde{h}}$  until killing time and output  $Z_{\zeta-}^{\tilde{h}}$

# A time-homogeneous generative process

Idealised algorithm:

1. Initialise  $Z_0^{\vec{h}} \sim \tilde{\beta} \approx \mathbb{P}_\alpha(Z_{\zeta-}^h)$ 
  - for ergodic forward process with stationary distribution  $\mu$  and small exponential killing rate  $r > 0$ , choose  $\tilde{\beta} = \mu$  [ $\leftrightarrow$  [ergodic diffusion model](#)]
  - for exponentially killed BM with small killing rate  $r > 0$ , choose  $\tilde{\beta} = \text{Laplace}(0, (2r)^{-1/2} \mathbb{I}_d)$  [ $\leftrightarrow$  [variance exploding diffusion model](#)]
  - for  $\kappa(dy) = \frac{1}{G_r(x_0, y)} \delta_z$ , choose  $\tilde{\beta} = \delta_z$
2. Simulate diffusion  $Z^{\vec{h}}$  until killing time and output  $Z_{\zeta-}^{\vec{h}}$

## Requirements for implementation

1. learn  $\nabla \log \vec{h}$  (only a function in space – no time component);

# A time-homogeneous generative process

Idealised algorithm:

1. Initialise  $Z_0^{\vec{h}} \sim \tilde{\beta} \approx \mathbb{P}_\alpha(Z_{\zeta-}^h)$ 
  - for ergodic forward process with stationary distribution  $\mu$  and small exponential killing rate  $r > 0$ , choose  $\tilde{\beta} = \mu$  [ $\leftrightarrow$  [ergodic diffusion model](#)]
  - for exponentially killed BM with small killing rate  $r > 0$ , choose  $\tilde{\beta} = \text{Laplace}(0, (2r)^{-1/2} \mathbb{I}_d)$  [ $\leftrightarrow$  [variance exploding diffusion model](#)]
  - for  $\kappa(dy) = \frac{1}{G_r(x_0, y)} \delta_z$ , choose  $\tilde{\beta} = \delta_z$
2. Simulate diffusion  $Z^{\vec{h}}$  until killing time and output  $Z_{\zeta-}^{\vec{h}}$

## Requirements for implementation

1. learn  $\nabla \log \vec{h}$  (only a function in space – no time component);
2. learn killing time  $\zeta$  of  $Z^{\vec{h}}$

# Learning to kill

## Polarity hypothesis

Assume that  $\mathcal{M} = \text{supp } \alpha$  is **polar** for  $X$ , i.e., for any  $x \in \mathbb{R}^d$ ,  $\mathbb{P}_x(\inf\{t > 0 : X_t \in \mathcal{M}\} < \infty) = 0$ .

# Learning to kill

## Polarity hypothesis

Assume that  $\mathcal{M} = \text{supp } \alpha$  is **polar** for  $X$ , i.e., for any  $x \in \mathbb{R}^d$ ,  $\mathbb{P}_x(\inf\{t > 0 : X_t \in \mathcal{M}\} < \infty) = 0$ .

## Theorem

Under the polarity hypothesis, the backward process  $Z^{\leftarrow h}$  is **killed at first entrance into  $\mathcal{M}$** .

# Learning to kill

## Polarity hypothesis

Assume that  $\mathcal{M} = \text{supp } \alpha$  is **polar** for  $X$ , i.e., for any  $x \in \mathbb{R}^d$ ,  $\mathbb{P}_x(\inf\{t > 0 : X_t \in \mathcal{M}\} < \infty) = 0$ .

## Theorem

Under the polarity hypothesis, the backward process  $Z^{\leftarrow h}$  is **killed at first entrance into  $\mathcal{M}$** .

Possible strategies to estimate a  $\delta$ -fattening  $\mathcal{M}_\delta = \{x : \text{dist}(x, \mathcal{M}) \leq \delta\}$  given data  $X^1, \dots, X^n \stackrel{\text{iid}}{\sim} \alpha$  and an estimator  $\hat{\mathfrak{z}}$  of  $\mathfrak{z} := \nabla \log \bar{h}$ :

- plug-in approach: estimate  $\mathcal{M}_\delta$  directly or indirectly by setting  $\widehat{\mathcal{M}}_\delta = (\widehat{\mathcal{M}})_\delta$ ; then set  $\hat{\zeta} := \inf\{t \geq 0 : Z_t^{\hat{\mathfrak{z}}} \in \widehat{\mathcal{M}}_\delta\}$
- use explosive behaviour of  $\mathfrak{z}$  as  $x \rightarrow \mathcal{M}$ :

## Theorem

Suppose that  $\mathcal{M}$  is polar for  $X$  and  $Y$  solving  $dY_t = \sigma(Y_t)dB_t$ . Then, it a.s. holds that

$$\zeta = \inf\left\{t \geq 0 : \sup_{s \leq t} |\mathfrak{z}(Z_s^{\leftarrow h})| = \infty\right\} = \inf\left\{t \geq 0 : \|\mathfrak{z}(Z^{\leftarrow h})\|_{L^2([0,t])} = \infty\right\}.$$



## Denoising score matching

- for  $\mathbb{P}_\alpha(Z_{\zeta-}^h \in \cdot)$ -a.e.  $x$

$$\begin{aligned}\mathfrak{s}(x) &= \nabla \log \tilde{h}(x) = \frac{1}{\tilde{h}(x)} \int \nabla_x G_r(x, y) \frac{1}{h(y)} \alpha(dy) = \int \nabla_x \log G_r(x, y) \frac{G_r(x, y)}{\tilde{h}(x)h(y)} \alpha(dy) \\ &= \mathbb{E}[\nabla_x \log G_r(x, Z_{\zeta-}^h) \mid Z_0^h = x] \\ &= \mathbb{E}_\alpha[\nabla_x \log G_r(x, Z_0^h) \mid Z_{\zeta-}^h = x]\end{aligned}$$

- this implies that on  $\mathbb{R}^d \setminus \mathcal{M}_\delta$ ,  $\mathfrak{s}$  agrees  $\mathbb{P}_\alpha(Z_{\zeta-}^h \in \cdot)$ -a.e. with the minimiser of

$$\mathcal{B}(\mathbb{R}^d; \mathbb{R}^d) \ni s \mapsto \mathbb{E}_\alpha \left[ \|s(Z_{\zeta-}^h) - \nabla \log G_r(Z_0^h, Z_{\zeta-}^h)\|^2 \mathbf{1}_{\{\|Z_{\zeta-}^h - Z_0^h\| > \delta\}} \right]$$

- note that if  $Z^h = Z$ , then  $\zeta \sim \text{Exp}(r)$  independent of  $Z$ ,  $Z_{\zeta-} = X_\zeta$  has full support and we have

$$\mathbb{E}_\alpha \left[ \|s(Z_{\zeta-}^h) - \nabla \log G_r(Z_0^h, Z_{\zeta-}^h)\|^2 \mathbf{1}_{\{\|Z_{\zeta-}^h - Z_0^h\| > \delta\}} \right] = r \mathbb{E}_\alpha \left[ \int_0^\zeta \|s(Z_t^h) - \nabla \log G_r(Z_0^h, Z_t^h)\|^2 \mathbf{1}_{\{\|Z_t^h - Z_0^h\| > \delta\}} dt \right]$$

## Projection learning

- we don't have to start the backward process approximately in  $\mathbb{P}_\alpha(Z_{\zeta_-}^h \in dy)$ : it will always be killed on the data support  $\mathcal{M}$  and different initialisations will yield different output distributions supported on  $\mathcal{M}$   $\rightsquigarrow$  **natural conditioning**
- a natural question is therefore what happens if we don't start the generative process from pure noise but something more informative, say a **masked** or **moderately noised** picture



- it turns out that the natural conditioning aspect entails a **blessing of dimensionality**

## Projection learning

Let  $Z$  be an **exponentially killed Brownian motion**. Then,

$$\tilde{h}(x) = \int G_r(x, y) \alpha(dy), \quad G_r(x, y) = 2(2\pi)^{-d/2} r \left( \frac{\sqrt{2r}}{|x - y|} \right)^{\frac{d-2}{2}} K_{\frac{d-2}{2}} \left( \frac{\sqrt{2r}}{|x - y|} \right).$$

For large  $d$ ,

$$\nabla \log \tilde{h}(x) \approx d \frac{\int \frac{y-x}{|y-x|^d} \alpha(dy)}{\int |y-x|^{2-d} \alpha(dy)}$$

and thus, if there is a unique projection  $x^* \in \arg \min_{y \in \mathcal{M}} |y - x|$  of  $x$  onto  $\mathcal{M}$ , then

$$\nabla \log \tilde{h}(x) \approx d \frac{x^* - x}{|x^* - x|^2} = d \frac{\text{sign}(x^* - x)}{|x^* - x|}$$

### Theorem

Let  $\delta, \varepsilon > 0$  and fix an observation  $x \in \mathbb{R}^d$ . If  $\alpha(B(x, r)) > \varepsilon$  for some ball  $B(x, r)$  with radius  $r > 0$  around  $y$ , then

$$\mathbb{P}\left(Z_{\zeta-}^{\tilde{h}} \in \mathcal{M} \cap B(x, (1+\delta)r) \mid Z_0^{\tilde{h}} = x\right) \geq 1 - \frac{1}{\varepsilon} (1+\delta)^{2-d}.$$

## Projection learning

Consider now estimators  $\hat{\mathfrak{s}}_n$ , an independent Brownian motion  $W$  and let  $\widehat{Z}^{\hat{\mathfrak{s}}_n}$  be the process solving

$$d\widehat{Z}_t^{\hat{\mathfrak{s}}_n} = \hat{\mathfrak{s}}_n(\widehat{Z}_t^{\hat{\mathfrak{s}}_n}) \mathbf{1}_{\{t \leq \hat{\zeta}\}} dt + \mathbf{1}_{\{t \leq \hat{\zeta}\}} dW_t, \quad \hat{\zeta} := \inf \left\{ t \geq 0 : \|\widehat{Z}^{\hat{\mathfrak{s}}_n}\|_{L^2[0,t]} > M \right\}.$$

### Theorem

Fix an observation  $x \in \mathbb{R}^d$ . Suppose that

- for any  $\tilde{\delta}, \delta, \varepsilon > 0$  it holds for sufficiently large  $n$  that

$$\mathbb{P} \left( \left\| (\hat{\mathfrak{s}}_n(Z^{\tilde{h}}) - \mathfrak{s}(Z^{\tilde{h}})) \mathbf{1}_{\{Z^{\tilde{h}} \notin \mathcal{M}_{\tilde{\delta}}\}} \right\|_{L^2(\zeta)} > \delta \mid Z_0^{\tilde{h}} = x \right) < \varepsilon$$

- for any  $n \in \mathbb{N}$  and  $\tilde{\delta} > 0$ , the function  $\hat{\mathfrak{s}}_n$  is  $L_{\tilde{\delta}}$ -Lipschitz on  $\mathcal{M}_{\tilde{\delta}}^c$

Let  $\delta, \varepsilon, \tilde{\delta}, \tilde{\varepsilon} > 0$ . If  $\alpha(B(x, r)) > \varepsilon$ , then, for sufficiently large  $M > 0$  and  $n \in \mathbb{N}$ ,

$$\mathbb{P}(\widehat{Z}_{\hat{\zeta}}^{\hat{\mathfrak{s}}_n} \in \mathcal{M}_{\tilde{\delta}} \cap B(x, (1 + \delta)r) \mid \widehat{Z}_0^{\hat{\mathfrak{s}}_n} = x) > 1 - \frac{1}{\varepsilon}(1 + \delta)^{2-d} - \tilde{\varepsilon}.$$

## Projection learning

Consider now estimators  $\hat{\mathbf{g}}_n$ , an independent Brownian motion  $W$  and let  $\widehat{Z}^{\hat{\mathbf{g}}_n}$  be the process solving

$$d\widehat{Z}_t^{\hat{\mathbf{g}}_n} = \hat{\mathbf{g}}_n(\widehat{Z}_t^{\hat{\mathbf{g}}_n}) \mathbf{1}_{\{t \leq \hat{\zeta}\}} dt + \mathbf{1}_{\{t \leq \hat{\zeta}\}} dW_t, \quad \hat{\zeta} := \inf \left\{ t \geq 0 : \|\widehat{Z}^{\hat{\mathbf{g}}_n}\|_{L^2[0,t]} > M \right\}.$$

### Theorem

Fix an observation  $x \in \mathbb{R}^d$ . Suppose that

- for any  $\tilde{\delta}, \delta, \varepsilon > 0$  it holds for sufficiently large  $n$  that

$$\mathbb{P} \left( \left\| (\hat{\mathbf{g}}_n(Z^{\tilde{h}}) - \mathbf{g}(Z^{\tilde{h}})) \mathbf{1}_{\{Z^{\tilde{h}} \notin \mathcal{M}_{\tilde{\delta}}\}} \right\|_{L^2(\zeta)} > \delta \mid Z_0^{\tilde{h}} = x \right) < \varepsilon$$

- for any  $n \in \mathbb{N}$  and  $\tilde{\delta} > 0$ , the function  $\hat{\mathbf{g}}_n$  is  $L_{\tilde{\delta}}$ -Lipschitz on  $\mathcal{M}_{\tilde{\delta}}^c$

Let  $\delta, \varepsilon, \tilde{\delta}, \tilde{\varepsilon} > 0$ . If  $\alpha(B(x, r)) > \varepsilon$ , then, for sufficiently large  $M > 0$  and  $n \in \mathbb{N}$ ,

$$\mathbb{P}(\widehat{Z}_{\hat{\zeta}}^{\hat{\mathbf{g}}_n} \in \mathcal{M}_{\tilde{\delta}} \cap B(x, (1 + \delta)r) \mid \widehat{Z}_0^{\hat{\mathbf{g}}_n} = x) > 1 - \frac{1}{\varepsilon} (1 + \delta)^{2-d} - \tilde{\varepsilon}.$$

Thank you for your attention!